

The Institutional Semantic Observatory

Engineering Information Ecosystems for Explainable AI

Edward Brash

School of Engineering and Computing
Christopher Newport University

July 2026

The Usual AI Picture Is Too Simple

Many people imagine an AI system as



That picture is useful, but incomplete.

For institutional reasoning, the hard part is often not the final answer. It is constructing the information environment in which reliable reasoning becomes possible.

What ISO Is

Institutional Semantic Observatory

ISO is an architecture for constructing, maintaining, observing, and reasoning over institutional knowledge.

It is not simply a chatbot and it is not simply a RAG system.

It is a measurement and reasoning framework for institutional information ecosystems.

The Core Question

Human reasoning benchmark

Given this information corpus, what would an intelligent human using appropriate reasoning skills conclude?

The goal is for the system to approximate that process:

- ① identify relevant evidence,
- ② organize it coherently,
- ③ reason transparently,
- ④ expose uncertainty,
- ⑤ and support a defensible human decision.

Why Human Reasoning Alone Is Not Enough

Scale

An expert may need evidence scattered across thousands of documents.

Even an excellent dean cannot read the entire corpus for every strategic question.

Recall

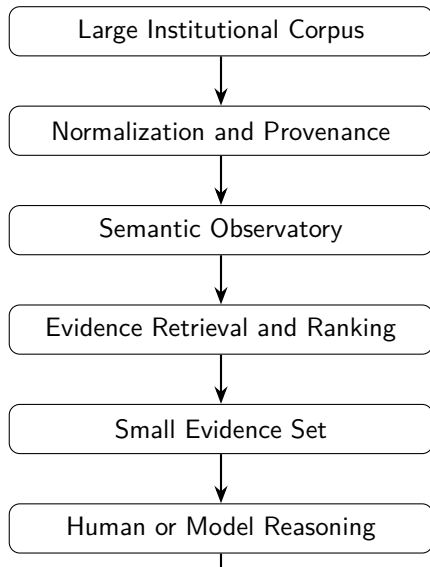
The right evidence may be hidden in ten documents among fifteen thousand.

The expert may reason well, but never see the crucial source.

Human reasoning is often not the bottleneck.

Evidence discovery is.

ISO Reframes the Bottleneck

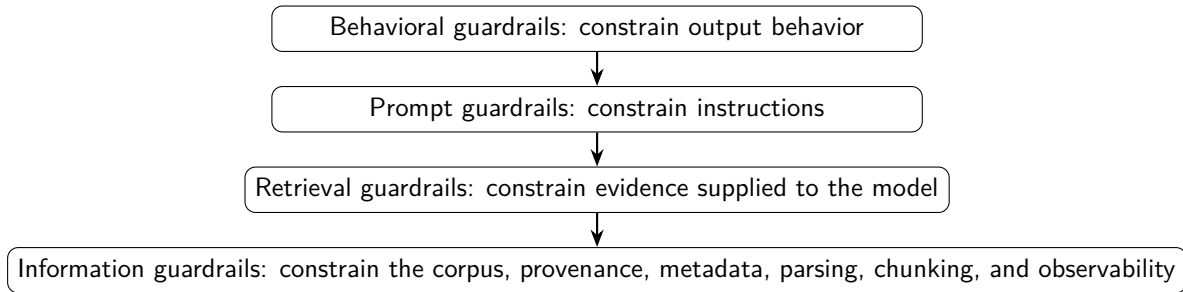


Two Philosophies of AI System Design

Conventional AI framing	ISO framing
Intelligence resides primarily in the model.	Intelligence emerges from the interaction between the reasoning engine and the information environment.
Build a very capable model, then prevent it from doing bad things.	Build an information environment in which good reasoning is the natural outcome.
Guardrails constrain the actor.	Guardrails engineer the environment.
Better model → better answer.	Better ecosystem → better evidence → better reasoning.

The most effective guardrail is a well-engineered information environment.

Guardrails: Where Do They Live?



ISO places many of the most important guardrails below the LLM, where they can be explicit, inspectable, and auditable.

Knowledge Objects Store Facts

Knowledge objects store facts.

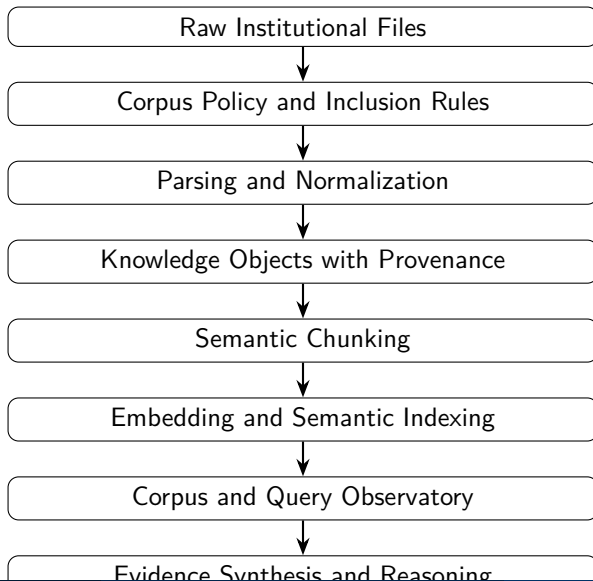
Documents, metadata, source paths, timestamps, parser information, chunk boundaries, and provenance.

Services derive meaning.

Retrieval, ranking, synthesis, decision briefs, observatory metrics, and scenario analysis.

This separation keeps the evidence layer stable while allowing multiple reasoning services to operate over it.

The ISO Pipeline



The Semantic Ecosystem

Working metaphor

An institutional corpus behaves less like a filing cabinet and more like an ecosystem.

Healthy semantic ecosystems have

- diversity,
- balance,
- provenance,
- redundancy without pathological duplication,
- observable structure,
- and resilience to noise.

Unhealthy ecosystems contain corrupted documents, dominant outliers, stale knowledge, duplicated artifacts, and isolated semantic islands.

The Observatory Is Not Decorative

Analogy

A physicist would not trust a detector analysis without detector monitoring, calibration checks, acceptance studies, and quality plots.

Likewise, we should not trust AI-assisted institutional reasoning without instrumentation of the underlying semantic environment.

The observatory is part of the scientific instrument.

Global metrics

- document count
- chunk count
- parser distribution
- file-type distribution
- largest documents
- chunk dominance
- Gini coefficient

Why it matters

- detects corrupted documents
- reveals duplicated material
- identifies dominant sources
- monitors pipeline health
- makes semantic risk visible

Local Information Density

A question lives somewhere in semantic space.

Some regions are densely supported by institutional evidence.

Other regions are sparse.

Key idea

Hallucination risk is often not merely a model problem. It may indicate low local information density near the query.

From Corpus Observatory to Query Observatory

Corpus Observatory

Measures the global health of the information ecosystem.

- corpus size
- dominance
- parser health
- distribution
- outliers

Query Observatory

Measures how well-supported a particular question is.

- local information density
- evidence diversity
- provenance quality
- retrieval stability
- evidence completeness

A Query Diagnostic Panel

Question

What resources would be required to launch a Mechanical Engineering major?

Local information density		9.4/10
Evidence diversity		8.8/10
Provenance quality		9.1/10
Retrieval stability		8.4/10
Evidence completeness		8.1/10

Expected reasoning reliability: **High**

When the Evidence Is Not Enough

Question

Should CNU eliminate the French major?

A good decision-support system should not invent enrollment, budget, faculty, or strategic evidence that is not present.

Instead it should report

- what evidence was retrieved,
- what conclusions that evidence supports,
- what conclusions it does not support,
- and what information is missing.

Epistemic humility is a feature, not a failure.

Diagnosing Hallucinations

Not all hallucinations have the same cause.

Failure mode	ISO response
Missing evidence	Improve corpus coverage; report insufficiency.
Poor retrieval	Improve retrieval, reranking, benchmarks, and diagnostics.
Ambiguous or conflicting evidence	Surface disagreement and uncertainty.
Reasoning error	Improve prompts, models, validation, or human review.

The symptom may be one wrong answer; the causes live at different layers of the system.

Evidence Quality Is Not Model Confidence

A language model may sound confident even when the evidence is weak.

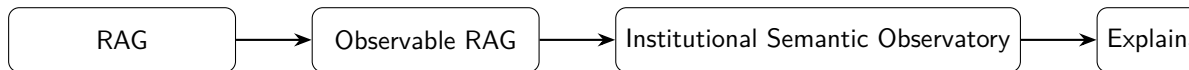
ISO question

How well-supported is this question by the available institutional evidence?

That is not the same as asking whether the model produced fluent text.

The observatory should measure evidence support before the final reasoning step.

From RAG to Institutional Reasoning



The progression is not primarily about larger models.

It is about better instrumentation of the information environment.

Research Questions

- Does local information density correlate with answer quality?
- Does retrieval stability predict benchmark performance?
- Does evidence diversity reduce reasoning errors?
- Can provenance quality predict user trust?
- Can evidence completeness identify when a system should refuse to recommend?
- Can query-level observatory metrics become calibrated reliability estimates?

These are empirical questions that can be tested inside a real institutional knowledge system.

Central claim

Reliable reasoning is not simply a property of the reasoning engine.

It is an emergent property of

- a well-engineered corpus,
- explicit provenance,
- observable retrieval,
- evidence diagnostics,
- and human-governed interpretation.

ISO is not mainly about making a model smarter.

It is about building the information environment in which good reasoning becomes possible.

Thank you.